

Address Mapping

- till DDR4, 1 channel of 64 bits

- DDR5 → dual channel

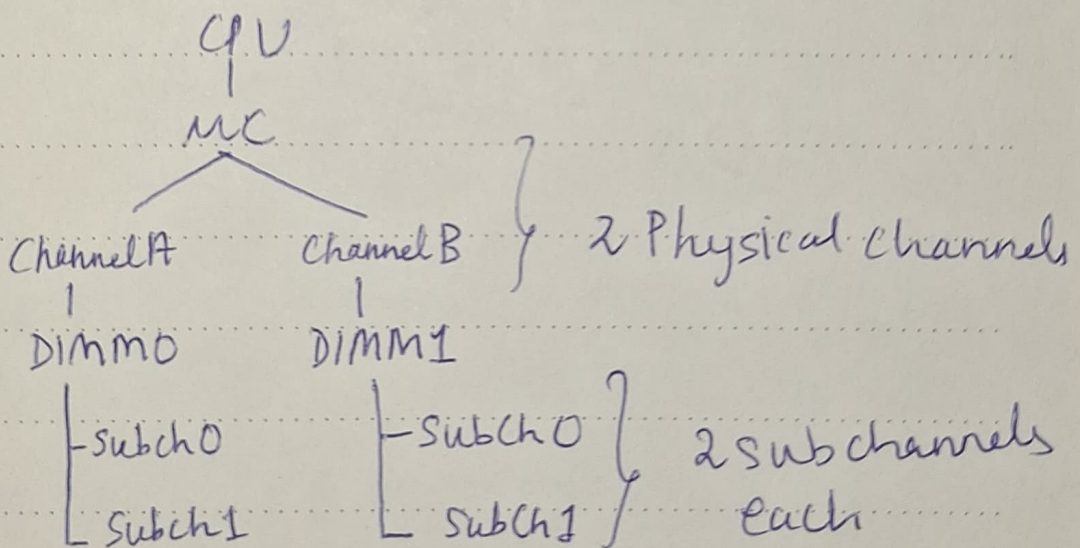
DIMM internally split into 2 independent channels.

Ch0 - 32 bits

Ch1 - 32 bits

improved efficiency

- subchannel has own ~~channel~~ ^{cmd} bus, addr bus, bank grps, bank



CPU Address



Channel, Rank, BG, Bank, Row, Channel

Setup time → no. of DIMMs

capacity

no. of ranks

timing parameters

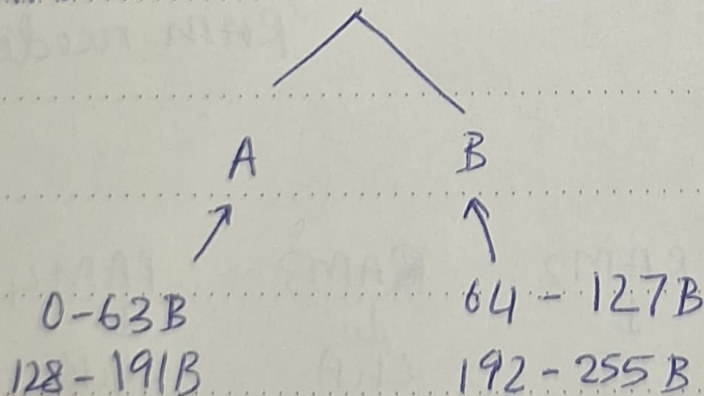
Topology

builds
address
mapping
tables

1) One RAM Stick → only if upgrading

16GB

1 DIMM ⇒ 2 subchannels



interleaving
(high BW)



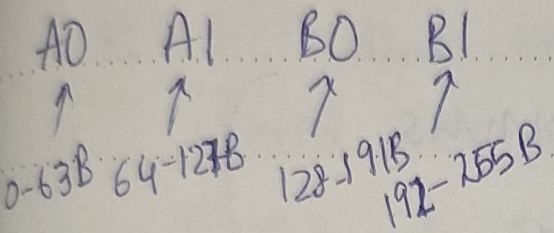
2x Two RAM Sticks → recommended

32GB

dual channel

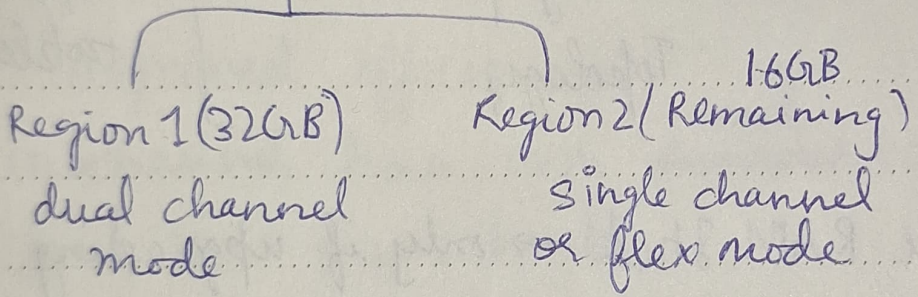
Interleaving

2x BW



3x Three RAM Sticks → avoid

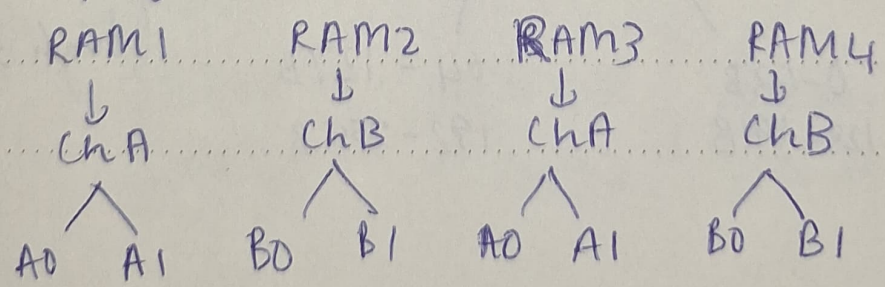
48GB



4x Four RAM Sticks → only use when lot of RAM needed

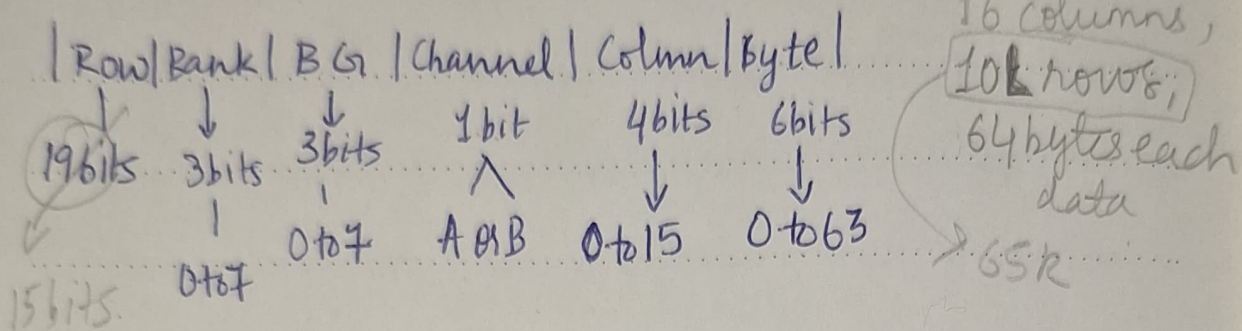
64GB

Mapping



36 → 32

• DDR5 Addr [36 bits]



(mapping can vary based on MC)

• Interleaving reduces stalls.

↳ parallel access

↳ ↑ BW

• During Boot what needs to be done.

no. of DIMMs, channels, ranks, subchannels, bank config, DIMM size, timing parameters, speed supported

DDR5 has 65k rows & 1-2k columns
16 bits 20 to 11 bits



8 BGr & 4 banks each
3 bits 2 bits